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Hit songs became wordier but not more meaningful, and literate societies grew faster: two tests of the conscious-language hypothesis

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Language is the medium through which a society stores what it has learned beyond the lifetime of any individual, and it has long been argued that the health of a society can be read from the health of its language. Two predictions follow from this view. First, societies whose members command their written language better should progress faster. Second, if modern listeners have lost access to meaning carried implicitly by melody, popular music should increasingly deliver meaning explicitly, through more words. Here we test both predictions with public data. Using test scores from 158 countries (1970 to 2015), literacy rates for 71 countries in about 1980, early modern European book production and real GDP per capita from the Maddison Project, we find that the breadth of literacy predicts faster growth over the following four decades (0.37 percentage points per year per 10 points of literacy), whereas the depth of measured skill predicts growth strongly in a long-run cross-section but not once its timing, schooling and geography are respected. Using three independent lyric collections covering the Billboard Year-End Hot 100 (1965 to 2020), we find that hit songs after 2010 contain about 60% more words than hits before 2000 (409 versus 267 words; Cohen's *d* = 1.08), replicated in all sources. The increase is not a post-2010 event, however: it began around 1990 and peaked in the 2000s. Only about a third of it reflects the rise of hip-hop, words per minute rose by 62% as songs shortened, and lyrics became markedly more repetitive. Hits now carry more words but less information per word, which is inconsistent with the stronger form of the conscious-meaning hypothesis.

Main

The proposition that a people's language and its fortunes rise and fall together is old, from the view that writing allows the experience of one generation to be inherited by the next without passing through the genes, to Orwell's warning [ref. 13] that the decline of a language and the decline of a polity reinforce one another. In a private philosophical notebook kept since 2019, the author framed this as the claim that written language is the first evidence of a communal, trans-generational "eternal self", and proposed two studies: one relating "the quality of a language expressing the eternal self" to the progress of the society that speaks it, and one comparing "lyrics per song before the 2000s and after the 2010s" to test whether, after 2010, listeners needed meaning translated into

explicit words rather than carried by melody. The notebook lists competing explanations for the second: social nonchalance, shorter attention spans, shorter notes, dopamine-seeking, and the rise of rap as poetry-led music.

Neither proposition can be tested in its metaphysical form. Both, however, make empirical predictions that public data can falsify. We operationalise the first as the claim that a society's command of its written language predicts its later material and health progress, and the second as the claim that the lyrical content of the most popular songs increased between the pre-2000 and post-2010 eras, with mechanism tests for each rival explanation. Both analyses followed an analysis plan fixed before outcomes were computed, and all deviations are reported.

Literacy and later progress

Linguists broadly agree that no natural language is intrinsically more expressive than another, even though language structure varies with the social structure of its speakers [ref. 12], so we did not rank languages. Instead we measured how well each society commands its written language along three dimensions: breadth [the share of adults who are literate], depth [harmonised test scores in reading, mathematics and science[1]] and historical written output [books consumed per capita in early modern Europe[3]]. Progress was measured as the annualised growth of real GDP per capita[2] and, secondarily, gains in life expectancy[8].

Breadth. Among 71 countries with literacy estimates near 1980, each 10 percentage points of literacy predicted 0.37 percentage points per year faster growth of GDP per capita between 1980 and 2018 [s.e. 0.12, $P = 0.002$], conditional on initial income [Fig. 1c and Table 1]. The association survived adjustment for average years of schooling [0.35, $P = 0.022$, $n = 65$] and continent fixed effects [0.31, $P = 0.004$]. Over 38 years, a 0.37-point annual difference compounds to about 15% higher income. Literacy did not predict gains in life expectancy over the same period [−0.07 years per 10 points, $P = 0.87$].

Depth. In a long-run cross-section of 107 countries, a one-standard-deviation higher average test score [83 points on the harmonised scale] was associated with 1.14 percentage points per year faster growth from 1970 to 2018 [s.e. 0.16, $P = 4 \times 10^{-13}$], and years of schooling added nothing once skills were included [Fig. 1a]. This reproduces the well-known finding that the quality of human capital, not its quantity, tracks long-run growth[4]. That specification, however, averages scores measured largely after growth had occurred, so it cannot separate cause from consequence. Our pre-specified test therefore used a country-by-five-year panel in which scores at time t predict growth over the following decade [357 observations, 128 countries]. Here one standard deviation of skill predicted 0.45 percentage points per year faster growth [95% confidence interval [CI] 0.07 to 0.83, $P = 0.020$], but the estimate fell to 0.35 [$P = 0.12$] when schooling was added, to 0.15 [$P = 0.56$] with continent fixed effects, and to −0.10 [$P = 0.81$] within countries over time [Fig. 1b]. Dropping Asia alone eliminated the effect [0.00]; dropping Africa strengthened it [0.48]. A reverse test found that past growth did not significantly predict later improvement in test scores [+1.3 points per percentage point of growth, $P = 0.20$], so the weak panel result is not simply growth raising scores. By our pre-registered decision rule, the depth hypothesis is not supported.

Historical written output. Across 11 European countries, book consumption per capita in 1475 to 1625 correlated

with GDP per capita in 1870 [Spearman ρ = 0.70, P = 0.016; Fig. 1d], with the Netherlands, Britain and Belgium at the top and Poland, Spain and Sweden at the bottom. It did not significantly predict growth from 1500 to 1800 [ρ = 0.37, P = 0.33, n = 9]. With so few units, this test is illustrative only.

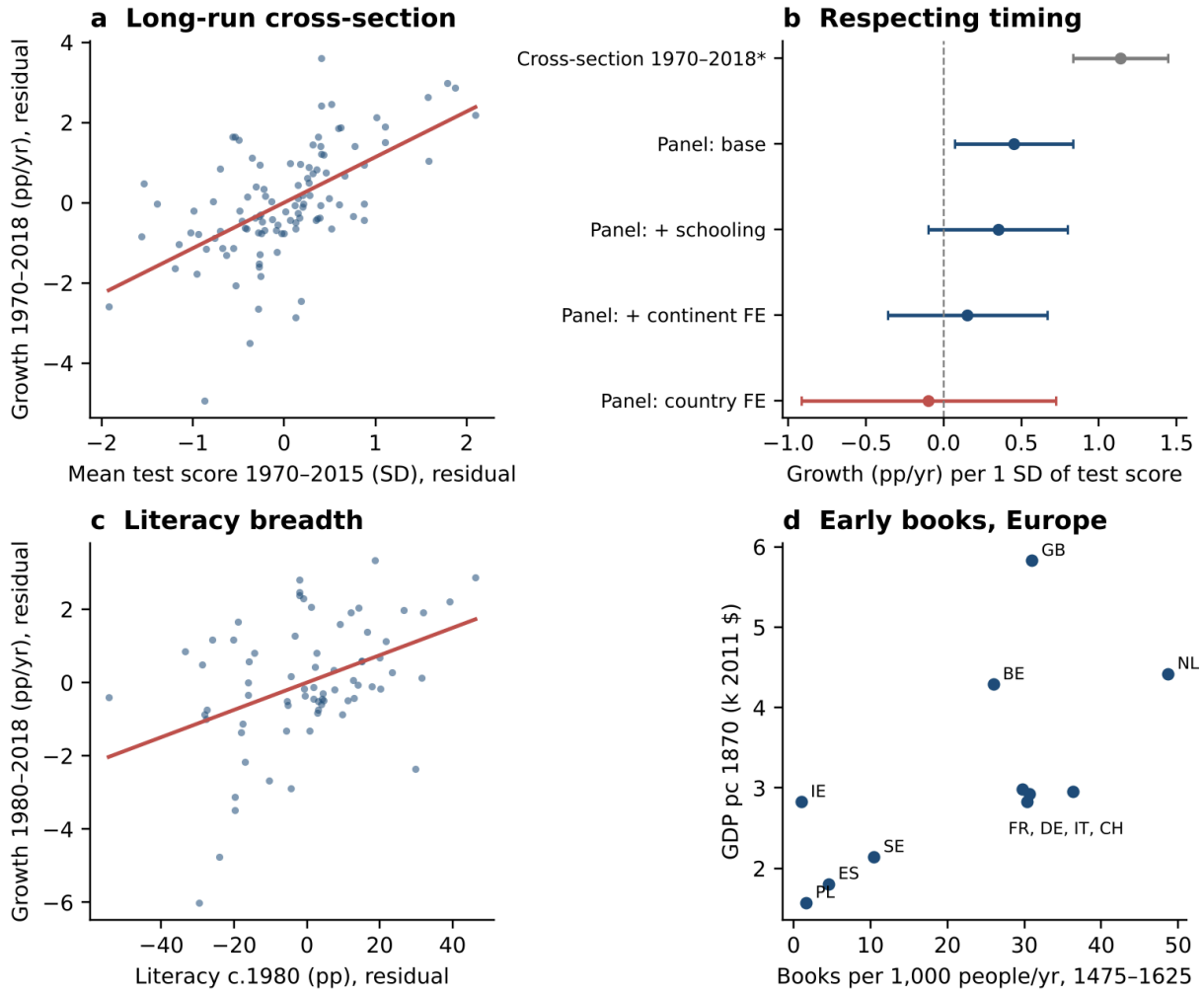


Fig. 1 | Command of written language and later economic progress. **a**, Long-run cross-section (exploratory): average harmonised test score, 1970 to 2015, against annualised growth of GDP per capita, 1970 to 2018, both residualised on log GDP per capita and years of schooling in 1970 (n = 107). **b**, Growth effect of a one-standard-deviation higher test score with 95% CIs. Grey, cross-section from **a**; blue, pre-registered panel in which scores at t predict growth from t to t + 10 (country-clustered errors), with successive controls; red, within-country estimate. **c**, Literacy rate near 1980 against growth from 1980 to 2018, both residualised on log GDP per capita in 1980 (n = 71). **d**, Book consumption per 1,000 inhabitants per year, 1475 to 1625, against GDP per capita in 1870, for 11 European countries. Lines are ordinary least-squares fits.

Table 1 | Study 1 estimates

Test	Language measure	Outcome	Estimate (s.e.)	*P*	*n*
S1-A base (panel)	Test score, per s.d.	Growth next 10 yr, pp/yr	0.45 (0.19)	0.020	357 / 128
S1-A + schooling	Test score, per s.d.	Growth next 10 yr	0.35 (0.23)	0.12	343 / 124
S1-A + continent FE	Test score, per s.d.	Growth next 10 yr	0.15 (0.26)	0.56	341 / 123
S1-A country FE	Test score, per s.d.	Growth next 10 yr	−0.10 (0.42)	0.81	343 / 124
S1-A reverse	Past growth, per pp/yr	Test-score change next 10 yr	1.29 (1.01)	0.20	295 / 96
S1-B	Literacy c.1980, per 10 pp	Growth 1980 to 2018	0.37 (0.12)	0.002	71
S1-B + schooling	Literacy, per 10 pp	Growth 1980 to 2018	0.35 (0.15)	0.022	65
S1-B + continent FE	Literacy, per 10 pp	Growth 1980 to 2018	0.31 (0.11)	0.004	71
S1-B life expectancy	Literacy, per 10 pp	Life-expectancy gain, years	−0.07 (0.42)	0.87	71
S1-C	Books per capita 1475 to 1625	GDP pc 1870 (Spearman ρ)	0.70	0.016	11
S1-D (exploratory)	Mean test score, per s.d.	Growth 1970 to 2018	1.14 (0.16)	4×10^{-13}	107
S1-D + continent FE	Mean test score, per s.d.	Growth 1970 to 2018	0.90 (0.18)	6×10^{-7}	107

All models control for log initial GDP per capita; panel models include period fixed effects. pp, percentage points; FE, fixed effects; *n* for panels is observations / countries.

Words per hit song

We counted the words in every Billboard Year-End Hot 100 song from three independently scraped lyric collections [Methods], using the same tokenizer for all. The two largest collections agreed closely on the songs they share [$r = 0.87$, $n = 4{,}595$; median ratio 1.00].

The primary prediction holds. Hits from 2010 to 2015 contained 409 words on average, against 267 for hits from

1965 to 1999: a difference of 142 words [95% CI 128 to 155], or +53%, Cohen's $d = 1.08$, $P < 10^{-50}$ [Fig. 2a and Table 2]. The second collection gave 412 against 278 words [+134, 95% CI 121 to 147, +48%] and the third 421 against 294 [+127, 95% CI 112 to 142, +43%]. Median song length rose from 243 to 385 words. Extending the post period to 2017 gave the same result [+144 words], and a fourth, Genius-based collection, calibrated on 674 overlapping songs, suggests the level held at about 430 to 446 words through 2020.

The timing does not match a post-2010 shift. Mean words per hit rose steadily through the decades: 186 in the 1960s, 230 in the 1970s, 259 in the 1980s, 352 in the 1990s, 430 in the 2000s and 412 in 2010 to 2017. Most of the increase therefore occurred during the 1990s and 2000s, and hits after 2010 were slightly shorter in words than hits of the 2000s [−18 words, 95% CI −33 to −3; Welch $P = 0.021$, Mann-Whitney $P = 0.31$]. Relative to the 1990s alone the post-2010 gain is modest [+59 words, +17%, $d = 0.37$].

Rap explains about a third. The share of hits tagged as hip-hop or rap rose from 5% before 2000 to 25% in 2010 to 2015 [Fig. 2b], and hip-hop hits carry roughly 240 more words than other hits. A twofold Oaxaca-Blinder decomposition[9,10] attributes 48 of the 135-word gap [36%] to this change in composition [17% to 41% depending on the reference weights]. The rest occurred within genres: songs not tagged as hip-hop rose from 255 to 375 words [+47%, $d = 1.16$; Fig. 2c], whereas hip-hop songs themselves did not become wordier [532 to 487, $P = 0.10$]. This is consistent with the notebook's view that rap reshaped popular music, but through its influence on every genre's delivery more than through its own share of the charts, in line with the timing of the hip-hop transition identified in the audio record[19].

Denser delivery, shorter songs, more repetition. The mechanism tests discriminate between the rival explanations [Table 2]. Songs became shorter [4.01 to 3.77 min, −6%], consistent with the attention-span account, but the shortening is small and the 1960s hits were shorter still [Fig. 2d]. Delivery became much denser: words per minute rose from 68 to 110 [+62%], and by 57% within non-hip-hop songs. This is the closest measurable analogue of the notebook's "shorter notes" hypothesis and is the largest effect we observed. Energy rose by 15% while tempo rose by only 2%, so songs became more intense without becoming faster. Crucially for the meaning hypothesis, distinct words rose by only 33% while total words rose by 53%, the compressibility of lyrics fell by 14% [more repeated material], and length-corrected vocabulary diversity fell slightly [moving-average type-token ratio −1.4%, $P = 0.028$]. The additional words are, to a large degree, repeated words, matching an independent analysis of 12,000 songs that found lyrics becoming simpler and more repetitive[7].

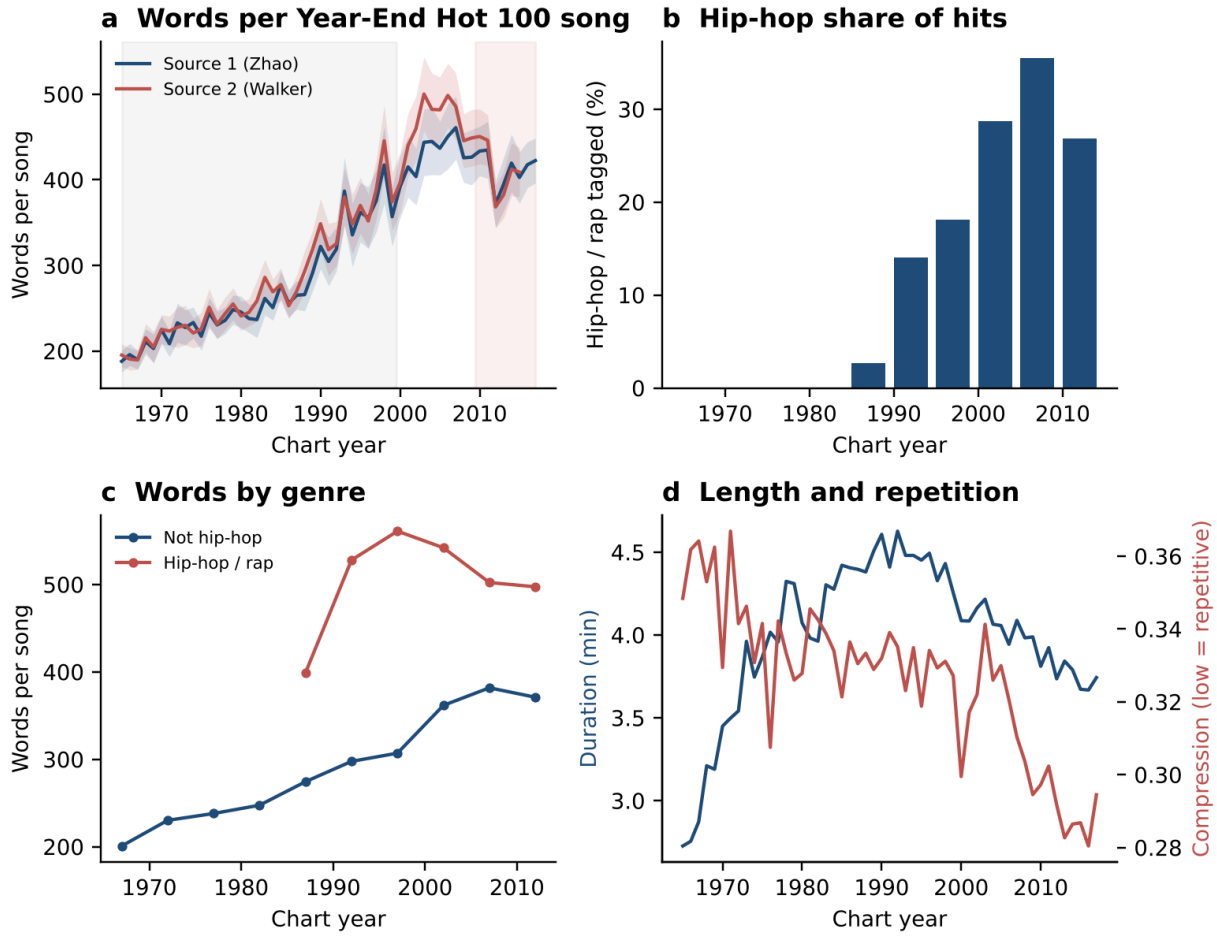


Fig. 2 | Words, genre, length and repetition in Billboard Year-End Hot 100 songs. a, Mean words per song by chart year in two independent lyric collections, with 95% CIs; grey and red bands mark the pre-2000 and post-2010 comparison windows. **b**, Share of hits tagged hip-hop or rap on Last.fm, five-year bins (tags available to 2015). **c**, Mean words per song for hip-hop and other hits (bins with ≥5 songs). **d**, Mean track duration (blue, left axis) and zlib compression ratio of the lyric text (red, right axis; lower values indicate more repetition).

Table 2 | Study 2: hits from 1965 to 1999 versus 2010 to 2017

Measure	Pre-2000	Post-2010	Change	95% CI of difference	Cohen's *d*
Words per song, source 1 (2010 to 2015)	267.4	409.1	+53%	+128 to +155	1.08
Words per song, source 2 (2010 to 2015)	277.9	411.7	+48%	+121 to +147	0.99
Words per song,	294.4	420.9	+43%	+112 to +142	0.92

Measure	Pre-2000	Post-2010	Change	95% CI of difference	Cohen's *d*
source 3 (2010 to 2015)					
Duration, min	4.01	3.77	−6%	−0.30 to −0.19	−0.25
Words per minute	67.8	109.6	+62%	+39 to +45	1.38
Distinct words per song	96.3	127.8	+33%	+27 to +36	0.69
Compression ratio (lower = more repetitive)	0.337	0.290	−14%	−0.053 to −0.041	−0.60
Moving-average type-token ratio	0.678	0.669	−1.4%	−0.018 to −0.001	−0.09
Energy (Spotify, 0 to 1)	0.601	0.692	+15%	+0.077 to +0.105	0.48
Tempo, BPM	119.1	121.9	+2%	+0.6 to +5.0	0.10
Hip-hop / rap share (tags, to 2015)	4.9%	25.3%	+20 pp		
Words, non-hip-hop songs only	255.5	375.0	+47%	+105 to +134	1.16
Words, hip-hop songs only	531.6	487.3	−8%	−97 to +7	−0.24

Rows from duration onwards use source 1 with Spotify audio features (pre n = 3,278; post n = 700); genre rows use the 2,571 source 1 songs matched to Last.fm tags. CIs from 10,000 bootstrap resamples. All differences except the last row have $P < 0.03$.

Discussion

Both studies return a qualified answer. For the first, a society's breadth of literacy is a robust predictor of its later economic growth, and early modern book culture marks the European economies that later became richest. The quality of skill measured by tests predicts growth only in specifications that cannot distinguish cause from effect; once timing, schooling and geography are respected, the estimate becomes small and uncertain. The defensible conclusion is narrower than the notebook's: what predicts progress is how many people can enter the written record, more than how finely the average person can use it, and nothing in these data ranks languages themselves.

None of these associations establishes causation; literacy may proxy for institutions, state capacity or cultural traits that independently drive growth.

For the second study, the core empirical claim is right: hit songs carry about half again as many words as they did before 2000, and the result replicates across three collections. The interpretation attached to it fares worse. The shift is concentrated in the 1990s and 2000s rather than after 2010, so it predates smartphones and the social-media environment that the nonchalance account invokes. More words did not mean more meaning in an information-theoretic sense: vocabulary grew far more slowly than word count, and repetition rose. Of the notebook's rival explanations, the data most strongly support the "shorter notes" account [denser, more percussive vocal delivery across all genres, reflected in a 62% rise in words per minute] and give partial support to the rap account [about one third of the increase by composition, plus diffusion of rap-style delivery into other genres]. The dopamine account is consistent with higher energy but cannot be tested with chart data, and the nonchalance and attention-span accounts receive at most weak support. A more accurate summary than "listeners need meaning spelled out" is that popular songs became rhythmic speech: more syllables per beat, repeated, delivered with more intensity.

Limitations are substantial. Chart data reflect commercial selection in one country, and since about 2014 streaming has changed what reaches the year-end chart. Lyric scrapes differ in how they record repeated choruses, which we addressed with three sources and a single tokenizer but cannot fully remove. Genre tags are assigned to artists, not songs. Test scores combine reading with mathematics and science, so they measure literacy only partially, and the panel covers mainly the period after 1990. Finally, "meaning" and the "eternal self" are not observable quantities; we tested measurable consequences that the notebook's arguments imply, and a failure of those consequences constrains the arguments without refuting their symbolic content.

Methods

Analysis plan

Hypotheses, operational definitions, comparison windows, primary tests and decision rules were written to a plan file before any outcome was computed, in the spirit of pre-registration[18]. Data were inspected only for coverage beforehand. Deviations are listed below.

Study 1 data

Real GDP per capita [2011 US dollars] and population were taken from the Maddison Project Database 2020[2]; harmonised learning outcomes [HLO], linking PISA, TIMSS, PIRLS and regional assessments on a common scale, for 158 countries at five-year intervals from 1970 to 2015, from Altinok, Angrist and Patrinos[1]; average years of schooling from Lee and Lee[5] and Barro and Lee[6], combined with UNDP data; adult literacy rates from the World Bank, UNESCO, CIA World Factbook and historical sources as compiled by Our World in Data; life expectancy from Riley[8], Clio Infra and the UN; and per-capita book consumption from Buringh and van Zanden[3]. All were obtained from the Our World in Data

datasets repository[17]. Regional aggregates were excluded; continents were assigned from ISO country codes.

Study 1 models

S1-A (primary): for each country and HLO year t (1970 to 2005), the outcome was 100 × [ln y_{t+10} − ln y_t]/10. Regressors were standardised HLO, ln y_t, years of schooling (nearest observation within five years) and period fixed effects, with standard errors clustered by country. Robustness models added continent or country fixed effects; leave-one-continent-out estimates were also computed. The reverse test regressed the ten-year change in HLO on growth over the preceding decade, controlling for HLO and ln y at t. S1-B: literacy in the observation nearest 1980 (1975 to 1985) predicting growth from 1980 to 2018 and change in life expectancy over the same period, with HC3 standard errors. S1-C: mean books per capita over the 1475 to 1625 observations against GDP per capita in 1870 and growth from 1500 to 1800, using Spearman correlations. S1-D (exploratory, added after the planned analyses): average HLO over all available years predicting growth from 1970 to 2018, controlling for 1970 income and schooling, following Hanushek and Woessmann[4].

Study 2 data

Source 1: Billboard Year-End Hot 100, 1960 to 2017, with scraped lyrics and Spotify audio features (duration, speechiness, energy, tempo)[14]. Source 2: Year-End Hot 100, 1965 to 2015, lyrics scraped independently[15]. Source 3: Year-End charts, 1950 to 2015, lyrics with Last.fm artist tags[16]. Source 4: Year-End Hot 100, 2010 to 2020, lyrics from Genius[16]. For each song we removed bracketed section labels, lower-cased the text and counted runs of letters and apostrophes. Songs with fewer than 20 tokens (instrumentals or failed scrapes) were excluded, leaving 5,485, 4,828, 4,027 and 1,021 songs. Lyrics were used only to compute these counts and are not reproduced.

Study 2 measures and tests

Windows were 1965 to 1999 (pre) and 2010 to 2015 (post, common to sources 1 to 3), with 2010 to 2017 as a pre-specified extension in source 1. Differences in means were tested with Welch's t-test and the Mann-Whitney U test, with 95% CIs from 10,000 bootstrap resamples and Cohen's d from the pooled standard deviation. Words per minute divided word count by Spotify duration (tracks shorter than 1 min or longer than 12 min excluded). Repetition was measured as the zlib compression ratio of the token string and as the moving-average type-token ratio with a 50-word window[11]. Source 4 was calibrated by the median ratio of its word counts to source 1 on 674 shared songs (1.043; r = 0.90). Hip-hop was defined as a Last.fm tag of rap, hip-hop or a named rap subgenre, matched to source 1 by title and year (2,571 songs). Decomposition used the pooled twofold Oaxaca-Blinder method[9,10], with pre- and post-period coefficients as bounds.

Deviations from the plan

[1] The plan defined rap as Spotify speechiness ≥ 0.33. That threshold classified only 0.7% of pre-2000 and 4.1% of post-2010 hits as speech-led, far below the known prevalence of hip-hop, so the primary decomposition uses Last.fm genre tags; the speechiness version is reported for completeness (it attributes 7% of the gap to composition). [2] S1-C used a ±20-year window to locate Maddison benchmarks near 1500 and 1800, and mapped Great

Britain to the United Kingdom series. [3] S1-D was added after the planned analyses and is labelled exploratory.

Data availability

All inputs are public: Our World in Data datasets repository (github.com/owid/owid-datasets); github.com/zhao1701/spotify-song-lyric-analysis; github.com/walkerkq/musiclyrics; github.com/kevinschaich/billboard; github.com/astromusicologist/Hidden-Hits.

Code availability

Analysis scripts (study1_language.py, study2_lyrics.py) and the analysis plan are available from the author on request.

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Author contributions

H.B. conceived both hypotheses and the rival explanations. Analyses were executed and drafted with AI assistance as described above.

Competing interests

The author declares no competing interests.